coal-creek-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))
```

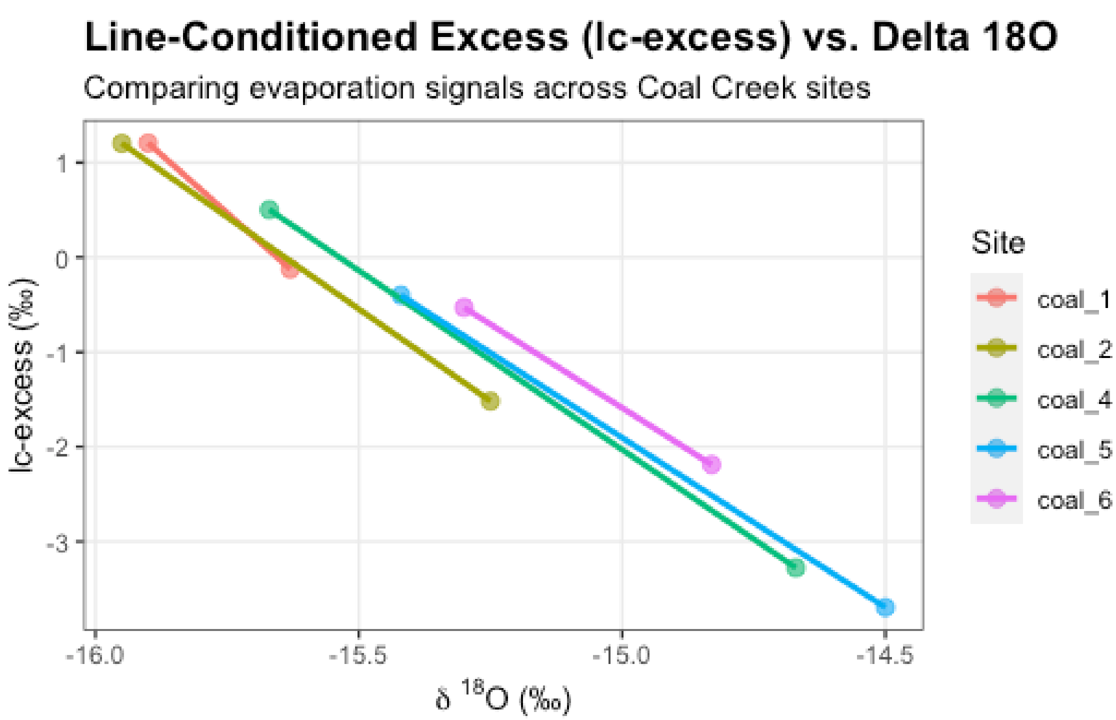
```
Rows: 5 Columns: 18
── Column
specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (12): site, location, cumulative_dams, Δdams, cumulative_dam_length, Δcu...
dbl  (4): total_dams, avg_dam_length, lc-excess_6-23, lc-excess_7-7
lgl  (2): normalized_stream_distance_m, Δx (m)

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
coal_creek <- compiled_isotope |> # the sites of focus
  filter(location %in% c("cc")) |>
```

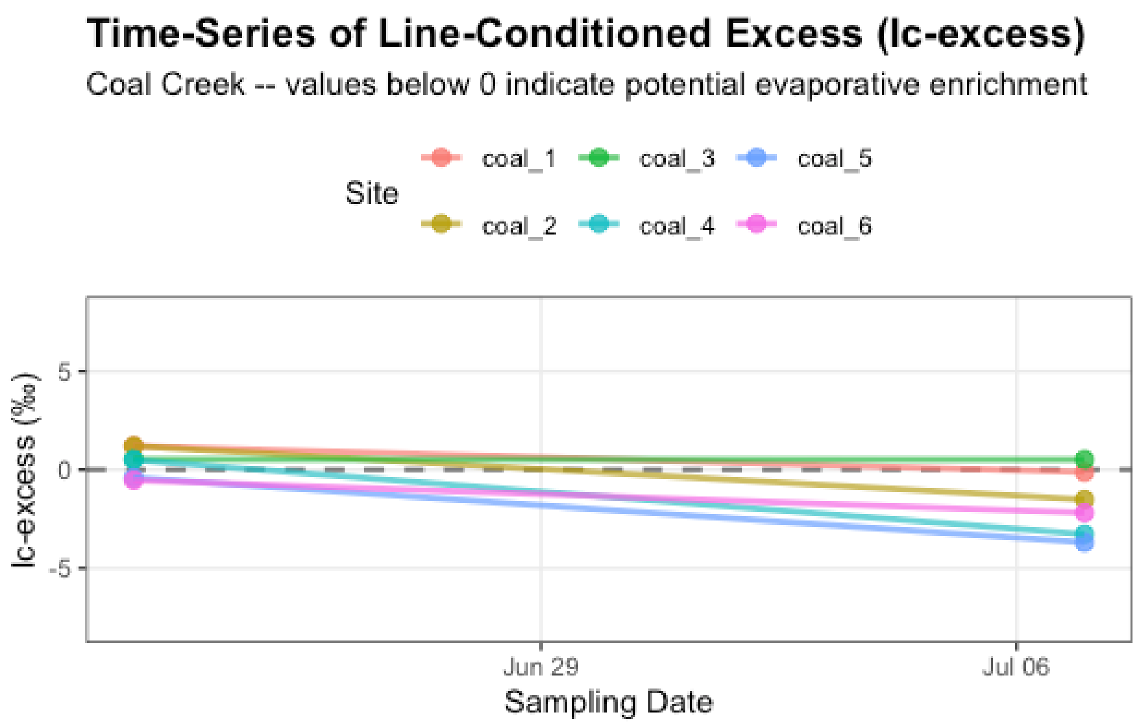
```
  filter(site != "coal_3")

coal_creek_all <- compiled_isotope |> # includes coal creek 3 that is a tributary
to main creek
 filter(location %in% c("cc"))
```

Figures

```
ggplot(data = coal_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
    subtitle = "Comparing evaporation signals across Coal Creek sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")),
    y = "lc-excess (\u2030)",
    color = "Site") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```

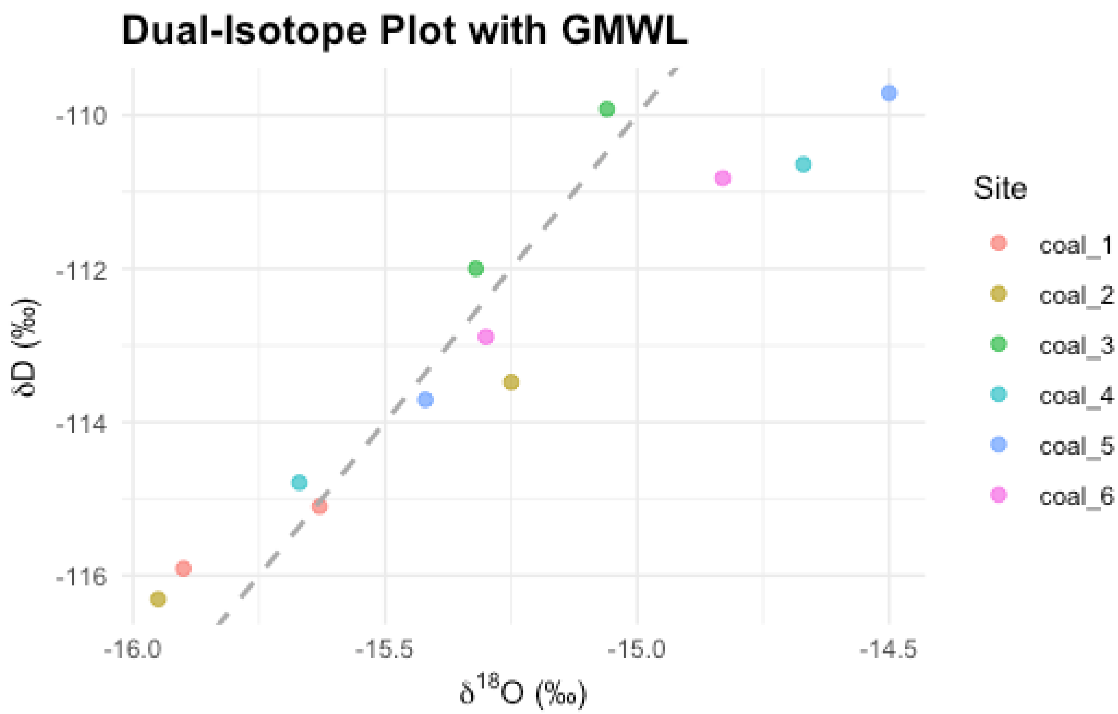


```
ggplot(data = coal_creek_all,
       aes(x = date, y = lc_excess, color = site, group = site)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", linewidth =
0.8) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_line(linewidth = 1, alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Coal Creek -- values below 0 indicate potential evaporative
enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8, 8) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



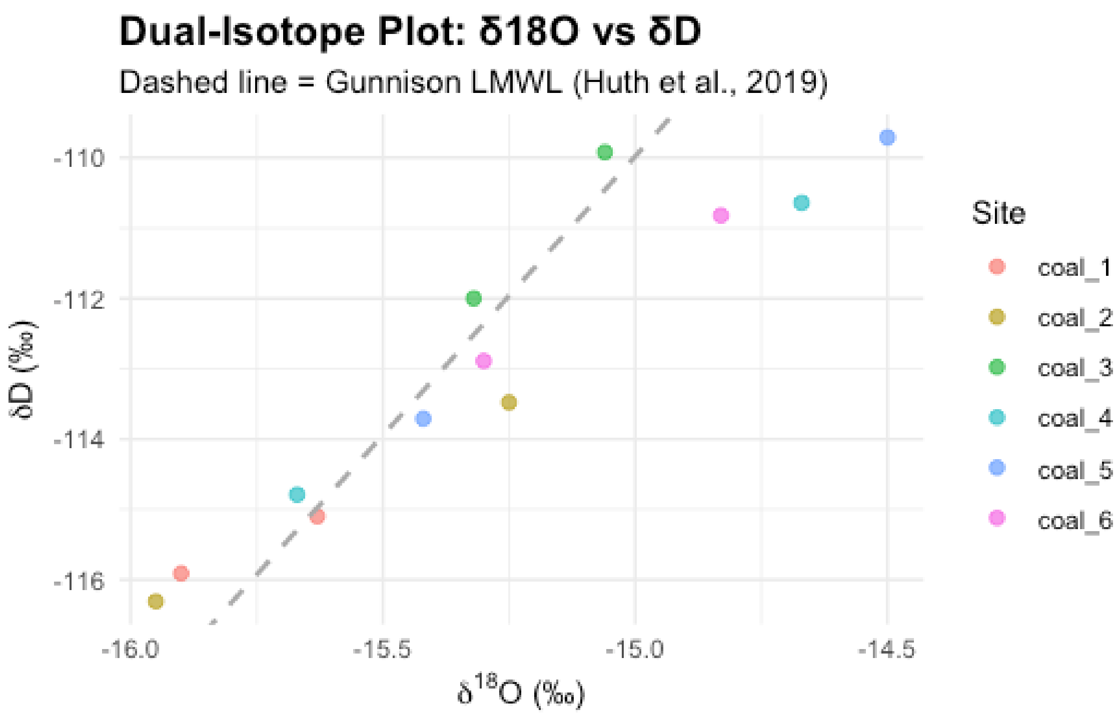

```
ggplot(coal_creek_all, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray",
```

```r
linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

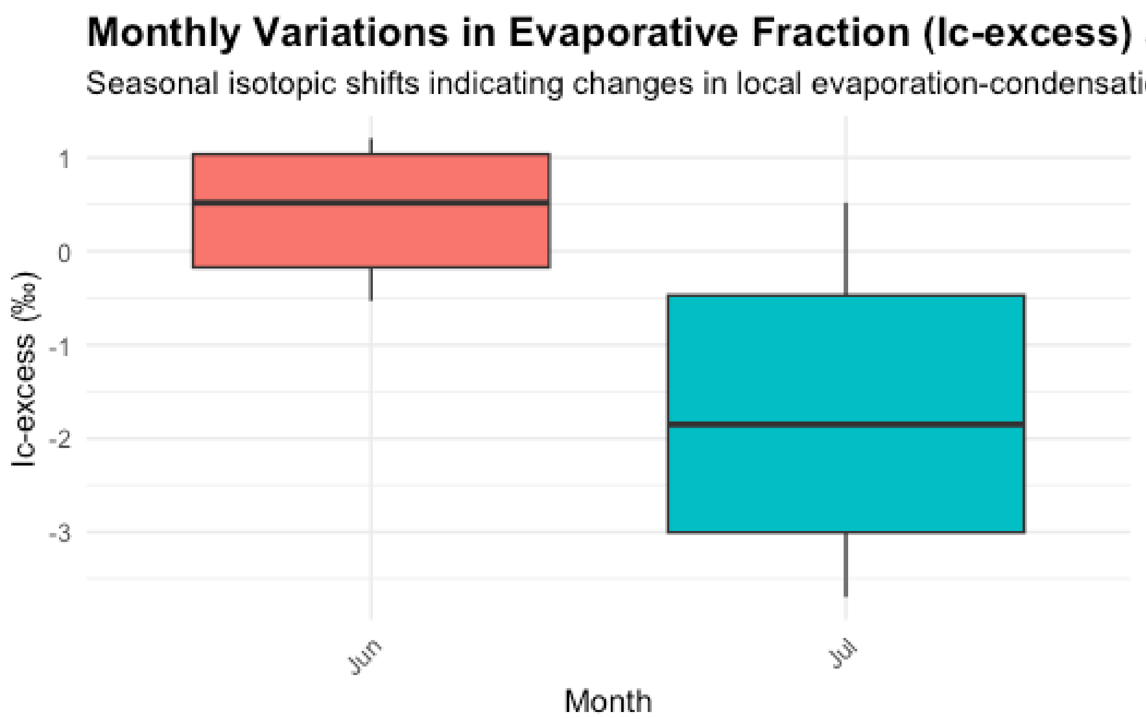



```r
# Coal Creek LMWL calculated from OPIC
lmwl_slope <- 7.93
lmwl_intercept <- 8.97

ggplot(coal_creek_all, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = lmwl_intercept, slope = lmwl_slope,
              linetype = "dashed", color = "darkgray", linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot: \u03b4180 vs \u03b4D",
    subtitle = "Dashed line = Gunnison LMWL (Huth et al., 2019)",
    x = expression(paste(delta^{18}, "O (\u2030)")),
```

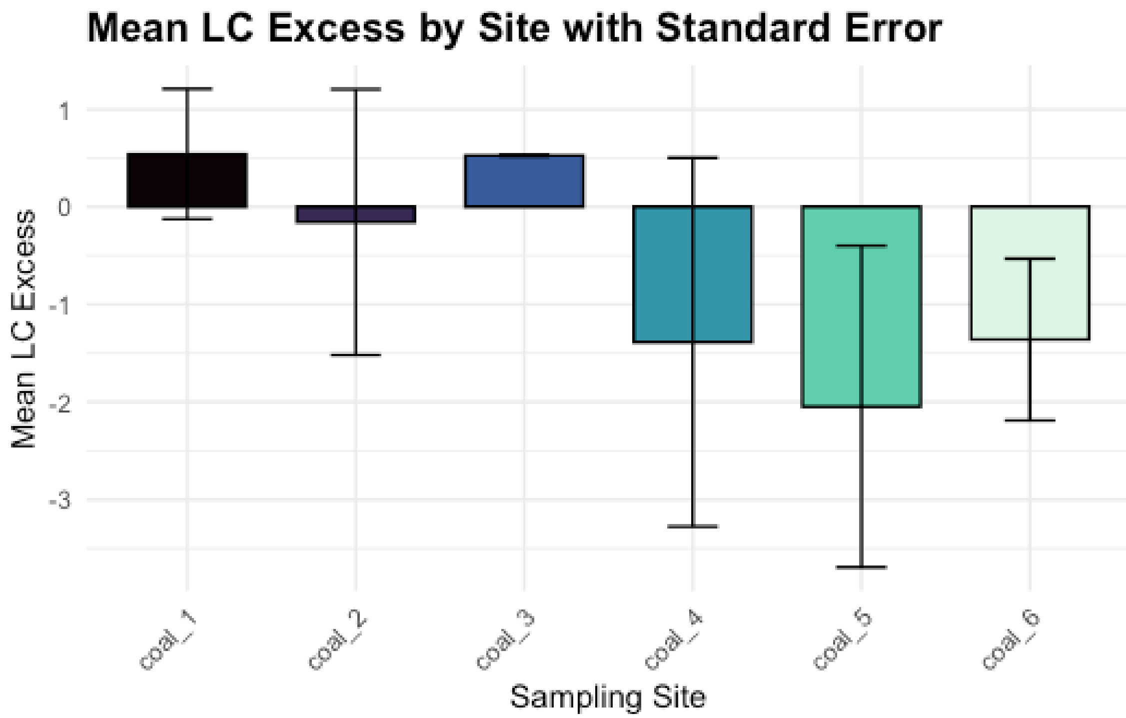
```
    y = expression(paste(delta, "D (\u2030)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



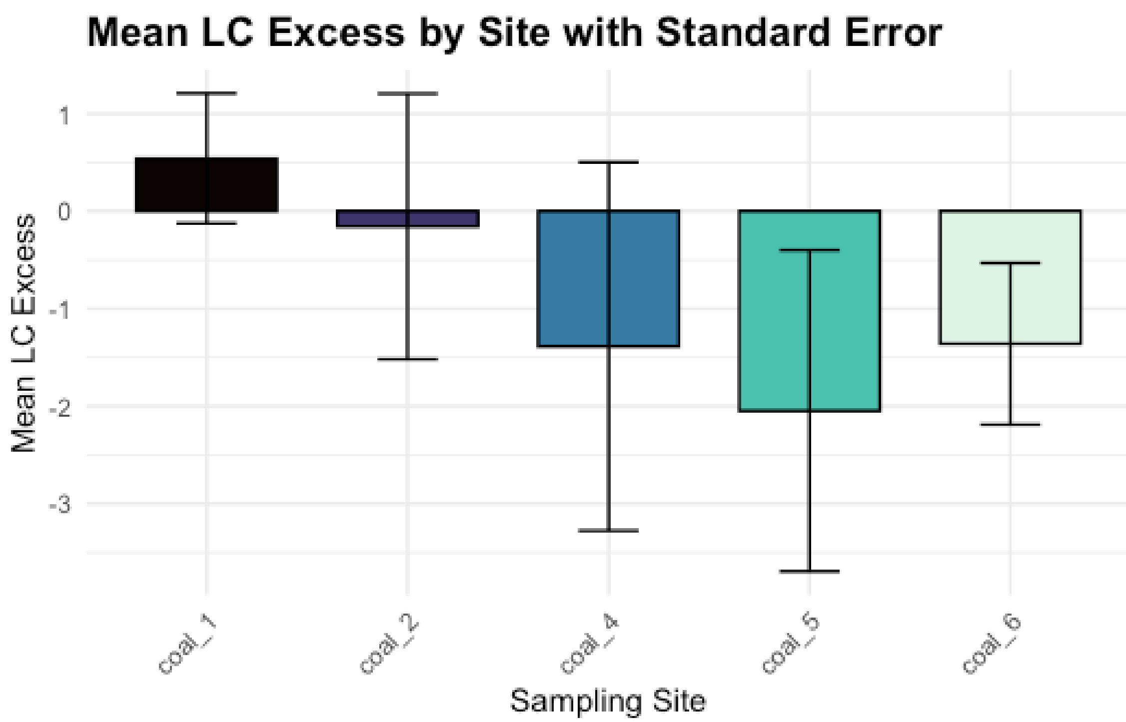

```
ggplot(coal_creek_all, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  labs(
   title = "Monthly Variations in Evaporative Fraction (lc-excess) at Coal Creek",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (\u2030)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
delta_summary_cc <- coal_creek_all |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_summary_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```




```r
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G") +
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```




```
gradient_cum_cc <- gradient_cc |>
  select(site, cumulative_dams, `Δlc-excess/Δavg_dam_length_6-23 (ppb/m)`,
         `Δlc-excess/Δcum_dam_length_7-7 (ppb/m)`) |>
  pivot_longer(
         cols   =  c(`Δlc-excess/Δavg_dam_length_6-23  (ppb/m)`,  `Δlc-excess/
Δcum_dam_length_7-7 (ppb/m)`),
    names_to = "date", values_to = "gradient") |>
   mutate(date = case_when(date == "Δlc-excess/Δavg_dam_length_6-23 (ppb/m)" ~
"2026-06-23",
                              date == "Δlc-excess/Δcum_dam_length_7-7 (ppb/m)"
~ "2026-07-07"),
                          gradient   =   as.numeric(gradient),   cumulative_dams   =
as.numeric(cumulative_dams),
         site = factor(site)) |>
  filter(!is.na(gradient))

ggplot(gradient_cum_cc, aes(x = site, y = gradient, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_hline(yintercept = 0, color = "black", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
    scale_fill_manual(values  =  wes_palette("FantasticFox1",  n  =  4,  type  =
"continuous")) +
```

```r
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_cc$cumulative_dams, gradient_cum_cc$site)
    lookup[x]
  }) +
  labs(
    x = "Cumulative dam count",
     y = expression(Delta * "lc-excess/" * Delta * "cumulative dam length (" *
"\u2030" * ")/m")
  ) +
  theme_bw(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold", size = 11),
    strip.background = element_rect(fill = "gray85", color = NA),
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    legend.title = element_text(face = "bold")
  )
```